

HW0428

2026-05-01

(a)

```
library(wooldridge)
data(mroz)

mroz_selected <- subset(mroz, inlf == 1)

mroz_selected$MOTHERCOLL <- ifelse(mroz_selected$motheduc > 12, 1, 0)
mroz_selected$FATHERCOLL <- ifelse(mroz_selected$fatheduc > 12, 1, 0)

mother_college_pct <- mean(mroz_selected$MOTHERCOLL, na.rm = TRUE) * 100
father_college_pct <- mean(mroz_selected$FATHERCOLL, na.rm = TRUE) * 100

nrow(mroz_selected)
```

```
## [1] 428
```

```
mother_college_pct
```

```
## [1] 12.14953
```

```
father_college_pct
```

```
## [1] 11.68224
```

參與勞動市場的樣本共有 428 人。母親受過部分大學教育（motheduc > 12）的比例約為 12.15%，父親受過部分大學教育（fatheduc > 12）的比例約為 11.68%。結果顯示，樣本中父母具有較高教育程度的比例並不高。

(b)

```
vars_for_corr <- mroz_selected[, c("educ", "MOTHERCOLL", "FATHERCOLL")]

cor_matrix <- cor(vars_for_corr, use = "complete.obs")

cor_matrix
```

```
##           educ MOTHERCOLL FATHERCOLL
## educ      1.0000000  0.3594705  0.3984962
## MOTHERCOLL 0.3594705  1.0000000  0.3545709
## FATHERCOLL 0.3984962  0.3545709  1.0000000
```

由相關係數矩陣可知，EDUC 與 MOTHERCOLL 的相關係數約為 0.3595，而 EDUC 與 FATHERCOLL 的相關係數約為 0.3985。兩者皆呈現正相關，表示父母教育程度越高，子女的受教育年數通常也較高。

(c)

```
library(ivreg)

wage_iv_model <- ivreg(
  log(wage) ~ educ + exper + I(exper^2) |
    MOTHERCOLL + exper + I(exper^2),
  data = mroz_selected
)

summary(wage_iv_model)

##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL +
##      exper + I(exper^2), data = mroz_selected)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.08719 -0.32444  0.04147  0.36634  2.35621
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.1327561  0.4965325  -0.267  0.78932
## educ         0.0760180  0.0394077   1.929  0.05440 .
## exper        0.0433444  0.0134135   3.231  0.00133 **
## I(exper^2)   -0.0008711  0.0004017  -2.169  0.03066 *
##
## Diagnostic tests:
##              df1 df2 statistic  p-value
## Weak instruments    1 424    63.563 1.46e-14 ***
## Wu-Hausman          1 423     0.742    0.39
## Sargan              0 NA         NA      NA
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6703 on 424 degrees of freedom
## Multiple R-Squared: 0.147,    Adjusted R-squared: 0.1409
## Wald test:    8.2 on 3 and 424 DF,  p-value: 2.569e-05
```

```
confint(wage_iv_model, level = 0.95)["educ", ]
```

```
##           2.5 %           97.5 %
## -0.001440867  0.153476785
```

以 MOTHERCOLL 作為 EDUC 的工具變數進行 IV 估計後，EDUC 的估計係數約為 0.0760，表示教育年數每增加一年，工資約增加 7.60%。此外，EDUC 的 95% 信賴區間為 [-0.0014, 0.1535]。由於信賴區間包含 0，因此在 5% 顯著水準下，無法認定 EDUC 對工資具有顯著影響

(d)

```
first_stage_d <- lm(
  educ ~ exper + I(exper^2) + MOTHERCOLL,
  data = mroz_selected
)
```

```
summary(first_stage_d)
```

```
##
## Call:
## lm(formula = educ ~ exper + I(exper^2) + MOTHERCOLL, data = mroz_selected)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.4267 -0.4826 -0.3731  1.0000  4.9353
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 12.079094   0.303118  39.849  < 2e-16 ***
## exper        0.056230   0.042101   1.336   0.182
## I(exper^2)   -0.001956   0.001256  -1.557   0.120
## MOTHERCOLL   2.517068   0.315713   7.973 1.46e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.133 on 424 degrees of freedom
## Multiple R-squared:  0.1347, Adjusted R-squared:  0.1285
## F-statistic: 21.99 on 3 and 424 DF,  p-value: 2.965e-13
```

```
t_val <- summary(first_stage_d)$coefficients["MOTHERCOLL", "t value"]
```

```
f_stat <- t_val^2
```

```
f_stat
```

```
## [1] 63.56307
```

第一階段迴歸結果顯示，MOTHERCOLL 的係數約為 2.5171，t 值約為 7.973，表示母親受過部分大學教育會顯著提高子女的教育年數。由於只有一個工具變數，因此 F-statistic 可由 t 值平方得到，約為 63.5631。此數值遠大於常用的弱工具變數判斷值 10，也高於 Stock-Yogo 臨界值 16.38，因此可認定 MOTHERCOLL 為強工具變數

(e)

```
wage_iv_model_ef <- ivreg(
  log(wage) ~ educ + exper + I(exper^2) |
  MOTHERCOLL + FATHERCOLL + exper + I(exper^2),
  data = mroz_selected
)
```

```
summary(wage_iv_model_ef)
```

```
##
## Call:
## ivreg(formula = log(wage) ~ educ + exper + I(exper^2) | MOTHERCOLL +
##       FATHERCOLL + exper + I(exper^2), data = mroz_selected)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.07797 -0.32128  0.03418  0.37648  2.36183
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.2790819   0.3922213  -0.712   0.47714
## educ         0.0878477   0.0307808   2.854   0.00453 **
## exper        0.0426761   0.0132950   3.210   0.00143 **
## I(exper^2)   -0.0008486   0.0003976  -2.135   0.03337 *
##
## Diagnostic tests:
##              df1 df2 statistic p-value
## Weak instruments    2 423    56.963  <2e-16 ***
## Wu-Hausman          1 423     0.519   0.472
## Sargan              1  NA     0.238   0.626
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6679 on 424 degrees of freedom
## Multiple R-Squared: 0.153,    Adjusted R-squared: 0.147
## Wald test: 9.724 on 3 and 424 DF,  p-value: 3.224e-06
```

```
confint(wage_iv_model_ef, level = 0.95)["educ", ]
```

```
##      2.5 %      97.5 %
## 0.02734575 0.14834956
```

同時使用 MOTHERCOLL 與 FATHERCOLL 作為工具變數後，EDUC 的估計係數約為 0.0878，表示教育年數每增加一年，工資約增加 8.78%。其 95% 信賴區間為 [0.0273, 0.1483]。由於信賴區間不包含 0，因此在 5% 顯著水準下，EDUC 對工資具有顯著正向影響。

(f)

```
library(car)
```

```
## 載入需要的套件：carData
```

```
first_stage_f <- lm(
  educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL,
  data = mroz_selected
)

linearHypothesis(
  first_stage_f,
  c("MOTHERCOLL", "FATHERCOLL")
)
```

```
##
## Linear hypothesis test:
## MOTHERCOLL = 0
## FATHERCOLL = 0
##
## Model 1: restricted model
## Model 2: educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL
##
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1      425 2219.2
## 2      423 1748.3   2    470.88 56.963 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
restricted_model <- lm(
  educ ~ exper + I(exper^2),
  data = mroz_selected
)

full_model <- lm(
  educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL,
  data = mroz_selected
)

anova(restricted_model, full_model)
```

```
## Analysis of Variance Table
##
## Model 1: educ ~ exper + I(exper^2)
## Model 2: educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1      425 2219.2
## 2      423 1748.3   2    470.88 56.963 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

聯合顯著性檢定結果顯示，MOTHERCOLL 與 FATHERCOLL 的聯合 F-statistic 約為 56.963，且 p-value 小於 0.001，表示兩個工具變數對 EDUC 具有顯著解釋力。此外，此 F 值大於常用判準值 10，也高於 Stock-Yogo 臨界值 19.93，因此可拒絕工具變數為弱工具變數的虛無假設。

(g)

```
mroz_e_clean <- na.omit(mroz_selected)

first_stage_e <- lm(
  educ ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL,
  data = mroz_e_clean
)

mroz_e_clean$educ_hat_e <- predict(first_stage_e)

iv_manual_e <- lm(
  log(wage) ~ educ_hat_e + exper + I(exper^2),
  data = mroz_e_clean
)

mroz_e_clean$residuals <- residuals(iv_manual_e)

sargan_reg <- lm(
  residuals ~ exper + I(exper^2) + MOTHERCOLL + FATHERCOLL,
  data = mroz_e_clean
)

r_squared <- summary(sargan_reg)$r.squared

n <- nrow(mroz_e_clean)

sargan_stat <- n * r_squared

p_value_sargan <- 1 - pchisq(sargan_stat, df = 1)

sargan_stat
```

```
## [1] 0.2136915
```

```
p_value_sargan
```

```
## [1] 0.6438893
```

Sargan test 的虛無假設為工具變數具有有效性，也就是工具變數與誤差項不相關。檢定結果顯示，Sargan test statistic 約為 0.2137，而 p-value 約為 0.6439。由於 p-value 大於 0.05，因此無法拒絕虛無假設，表示目前沒有足夠證據認為 MOTHERCOLL 與 FATHERCOLL 為無效工具變數。